

You must show work to receive credit. Do not use a calculator/computer for problems with a ★. Each problem is of equal value.

Use the given conditions to write an equation for the line in slope-intercept form, if possible.

1. Slope = -3 , passing through $(6, 6)$.

2. Slope = 4 passing through $(7, 3)$.

Find the equation of the least squares line.

3. The paired data below consists of the test scores of 6 randomly selected students and the number of hours they studied for the test.

Hours (x)	5	10	4	6	10	9
Score (y)	64	86	69	86	59	87

Write a cost function for the problem. Assume that the relationship is linear.

4. A moving firm charges a flat fee of \$40 plus \$35 per hour. Let $C(x)$ be the cost in dollars of using the moving firm for x hours.

Solve the system of two equations in two unknowns.

5. ★
$$\begin{aligned} x + y &= -2 \\ x - y &= 18 \end{aligned}$$

Solve the system of two equations in two unknowns.

6. ★
$$\begin{aligned} x + 6y &= -5 \\ -4x + 7y &= 20 \end{aligned}$$

Solve the problem.

7. A digital pen manufacturer has decided to come out with a new and affordable pen. The fixed cost for the production of this new pen line is \$16,600 and the variable costs are \$68 per pen. The company expects to sell the pens for \$159 a piece. Formulate a function $C(x)$ for the total cost of producing x new pens and a function $R(x)$ for the total revenue generated by the sales of x pens.

8. There are 600 people at a concert. The admission prices was \$10 for adults and \$5 for children. The admission receipts were \$4650. How many adults and how many children attended?

9. Sweet Honey Bakery serves desserts. Two servings of yogurt and three servings of apple cobbler has 1670 calories. One serving of yogurt and two servings of apple cobbler has 1030 calories. Find the caloric content of each item.

10. After two years on the job, a therapists salary was \$60,000. After seven years on the job, her salary was \$72,500. Let y represent her salary after x years on the job. Assuming that a change in her salary over time can be approximated by a straight line, give an equation for this line in the form $y = mx + b$.

11. FedUps delivers packages which cost \$3.00 per package to deliver. The fixed cost to run the delivery truck is \$250 per day. If the company charges \$6.00 per package, how many packages must be delivered to make a profit of \$131?

12. A shoe company will make a new design of walking shoe. The fixed cost for the production will be \$24,000. The variable cost will be \$31 per pair of shoes. The shoes will sell for \$106 for each pair. How many pairs of shoes will have to be sold for the company to break even on this new line of shoes?

13. Let the supply and demand functions for gourmet hot chocolate packs be given by

$$p = S(q) = \frac{9}{4}q \text{ and } p = D(q) = 70 - \frac{5}{4}q$$

where p is the price in dollars and q is the number of batches. Find the equilibrium quantity and the equilibrium price.

Find the slope of the line passing through the given pair of points .

14. $(3, 4)$ and $(3, -8)$

15. $(-4, 1)$ and $(-9, 1)$

16. $(5, -5)$ and $(-7, -3)$

17. $(6, 5)$ and $(-4, 5)$

The sizes of two matrices A and B are given. Find the sizes of the product AB and the product BA , whenever these products exist.

18. A is 1×3 , and B is 3×1 .

a. AB :

b. BA :

Find the matrix if possible.

19. ★ $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ -2 & 1 \\ 0 & 3 \end{bmatrix}$

20. ★ $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ -2 & 1 \end{bmatrix}$

Solve the problem.

21. ★ Let $A = \begin{bmatrix} -1 & 0 \\ 6 & 3 \end{bmatrix}$ Find $-2A$.

22. ★ Let $A = \begin{bmatrix} -1 & 0 \\ 6 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 6 \\ 3 & 1 \end{bmatrix}$. Find $A - B$.

23. ★ Let $A = \begin{bmatrix} -1 & 0 \\ 6 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 6 \\ 3 & 1 \end{bmatrix}$. Find $A + B$.

24. ★ Let $A = \begin{bmatrix} -1 & 0 \\ 6 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 6 \\ 3 & 1 \end{bmatrix}$. Find $2A + B$.

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

25. In one or more paragraphs, explain what it means to “break even” in business.